

# TD Series Servo CANopen Protocol



Version number

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## Chapter One Summary

### 1.1 CAN summary

The two control systems are a high-level protocol based on the CAN (control LAN) serial bus system and CAL (CAN application layer). The communication protocol assumes that the hardware of the connected device has a ISO 11898 compliant CAN transceiver and a CAN controller. Use the CAN 2.0A uniformly

CAN ,2.0A: supports the standard frame, defines the 11-bit identification area.

CAN, 2.0B: Support for remote frame, defined 29bit identification area.

The physical medium used by the controller is a double-line bus that meets the ISO 11898 standard and adopts a split-drive mechanism and public feedback. The maximum length of the bus depends on the communication speed, as specified below

| Communication baud rate | Maximum bus length |
|-------------------------|--------------------|
| 1M bit/s                | 25 m               |
| 500k bit/s              | 100 m              |
| 250k bit/s              | 250 m              |
| 125k bit/s              | 500 m              |
| 100k bit/s              | 600 m              |
| 50k bit/s               | 1000 m             |

Theoretically, there can be up to 127 nodes on the bus. However, in practice, the maximum number of nodes depends on the performance of the CAN transceiver used. Each node is an independent, device, and for the servo drive, each logical device corresponds to an axis under its control

### 1.2 Overview of the communication interface and the object dictionary

The communication interface mainly defines the protocol of how to transmit the process data message (PDO), service data message (SDO), network status management message (NMT) and other related messages.

The core concept of CANopen is the device object dictionary (OD: Object Dictionary), and CANopen communication can access all the parameters of the drive through the object dictionary (OD). Note: The object dictionary is implemented in the CANopen. Objects in the object dictionary

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can be identified by a known 16-bit index and an 8-bit index (see the list of object dictionaries in the appendix). CANopen has classified the object dictionary in the table below:

| Main index  | Target                                   | Main index  | Target                                               |
|-------------|------------------------------------------|-------------|------------------------------------------------------|
| 0000H       | Reserve                                  | 2000H~5FFFH | Manufacturers-specific objects                       |
| 0001H~001FH | Static data type                         | 6000H~67FFH | The object of the logical device 1                   |
| 0020H~003FH | Complex data type                        | 6800H~6FFFH | The object of the logical device 2                   |
| 0040H~005FH | Manufacturers-specific data type         | 7000H~77FFH | The object of the logical device 3                   |
| 0060H~007FH | Equipment specification: basic data type | ***         | ***                                                  |
| 0080H~009FH | Equipment specification: basic data type | 9800H~9FFFH | Objects for the logical device 8                     |
| 00A0H~0FFFH | Reserve                                  | A000H~BFFFH | Objects used by the standard interface specification |
| 1000H~1FFFH | Communication specification              | C000H~FFFFH | Reserve                                              |

One of the most commonly used objects is

- 1) 1000H-1 FFFH, the object used in the communication specification. Reading and writing these objects can realize the configuration of the network, PDO mapping and other operations.
- 2) 2000H-5 FFFH, manufacturer-specific object. Read and write to the Pnxxx parameter can be implemented through these objects.
- 3) 6000H-9 FFFH, the object of the logical device 1-8.belongs to the objects in the CIA 402

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device subprotocol, operation on these objects can realize various operation modes of the servo drive. You can also find the index of the relevant objects through the object dictionary list in the Appendix.

## Chapter Two CANOPEN Communication

CANopen The communication model defines the following messages (communication objects):

| Abbreviation | Detailed said           | Illustration                                                                                       |
|--------------|-------------------------|----------------------------------------------------------------------------------------------------|
| SDO          | Service Data Object     | For non-time-critical data, such as, for example, parameters.                                      |
| PDO          | Process Data Object     | Used to transfer time-critical process data (given value, control word, status information, etc.). |
| SYNC         | Synchronization Message | Used to synchronize the CAN nodes.                                                                 |
| EMCY         | Emergency Message       | Alarm events used for transmitting the drive.                                                      |
| NMT          | Network Management      | For the CANopen network management.                                                                |
| Heartbeat    | Error Control Protocol  | Used to monitor the lifetime status of all the nodes.                                              |

The communication format adopts the standard-frame mode. Where the COB-ID (communication object identifier) is assigned:

| FC |   |   |   | Motor ID (node address) |   |   |   |   |   |   |
|----|---|---|---|-------------------------|---|---|---|---|---|---|
| 10 | 9 | 8 | 7 | 6                       | 5 | 4 | 3 | 2 | 1 | 0 |

### 2.1 Service data object SDO

The dictionary of objects that the SDO uses to access a device. The visitor is called the customer (client), and the CANopen device to which the object dictionary is accessed and provides the requested service is called the server (server). Customer CAN message and server response CAN message always contain 8 bytes of data (although not all bytes of data must make sense). A customer's request must have a response from the server.

**The basic SDO structure is as follows**

| Frame ID | Len | Byte0        | Byte 1~2     | Byte 3           | Byte 4~7 |
|----------|-----|--------------|--------------|------------------|----------|
|          | 8   | CS character | Object index | Object sub-index | Data     |

#### Frame ID

|                                                      |
|------------------------------------------------------|
| Frame ID that sends the SDO data= 0x600+\$NODEID     |
| Frame ID that receives the SDO data= 0x 580+\$NODEID |

#### CS command character

| Read operation |             |    | Index | Subindex | Data        |
|----------------|-------------|----|-------|----------|-------------|
| Command        | Read        | 40 | XX XX | XX       | 00 00 00 00 |
| Answer         | unit 8/int8 | 4F | XX XX | XX       | XX XX XX XX |

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|                           |                    |    |                                                 |          |               |
|---------------------------|--------------------|----|-------------------------------------------------|----------|---------------|
|                           | uint16/int16       | 4B | XX XX                                           | XX       | XX XX XX XX   |
|                           | uint32/int32       | 4B | XX XX                                           | XX       | XX XX XX XX   |
| Write operation           |                    |    | Index                                           | Subindex | Data          |
| Command                   | uint8/int8         | 2F | XX XX                                           | XX       | XX XX XX XX   |
|                           | uint16/int16       | 2B | XX XX                                           | XX       | XX XX XX XX   |
|                           | uint32/int32       | 23 | XX XX                                           | XX       | XX XX XX XX   |
| Answer                    | success            | 60 | XX XX                                           | XX       | XX XX XX XX   |
|                           | exception response | 80 | XX XX                                           | XX       | F0 F1 F2 F3   |
| Error code<br>F3 F2 F1 F0 |                    |    | Description Use the source program error report |          |               |
|                           |                    |    |                                                 |          | To be written |
|                           |                    |    |                                                 |          |               |

**Index** Reference to the index supported by the index table

**Sub-index** The branch of the index

**Data value** High bytes in front, low bytes in behind

### 2.2 Process data object PDO

The PDO is used to transmit real-time data from one producer to one or more consumers. Data transfer is limited to 1 to 8 bytes. PDO communications has no protocol constraints (meaning that the data content is pre-defined), so consumers can process the received data in a very short period of time. The PDO data content is defined only by its CAN ID, assuming that producers and consumers are aware of the data content of this PDO.

|                               | Index                                |
|-------------------------------|--------------------------------------|
| RPDO Communication parameters | 1400 <sub>h</sub> ~15FF <sub>h</sub> |
| RPDO Mapping parameters       | 1600 <sub>h</sub> ~17FF <sub>h</sub> |
| TPDO Communication parameters | 1800 <sub>h</sub> ~19FF <sub>h</sub> |
| TPDO Mapping parameters       | 1A00 <sub>h</sub> ~1BFF <sub>h</sub> |

Each PDO is described in 2 objects in the object dictionary:

PDO communication parameters: include the COB-ID to be used by PDO, transmission type, forbidden time and timer period.

PDO mapping parameters: It contains a list of objects in an object dictionary that are mapped into the PDO, including their bit data length (in bits). Producers and consumers must know this mapping to explain the PDO content.

This drive supports 4 dynamic PDO maps. Using SDO message, you can configure PDO mapping parameters, PDO and communication parameters. Only the detailed communication parameters



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and mapping parameters of sending and receiving PDO are listed below. The other 3 routes send / accept the same detailed communication parameters and mapping parameters of the PDO

### TPDO channel configuration

|          |                      |                             |                      |                  |
|----------|----------------------|-----------------------------|----------------------|------------------|
| index    | 1800 <sub>h</sub>    | TPDO1channelconfiguration   | COB-ID               | 180 <sub>H</sub> |
|          | 1801 <sub>h</sub>    | TPDO2channelconfiguration   | COB-ID               | 280 <sub>H</sub> |
|          | 1802 <sub>h</sub>    | TPDO3 channel configuration | COB-ID               | 380 <sub>H</sub> |
|          | 1803 <sub>h</sub>    | TPDO4 channel configuration | COB-ID               | 480 <sub>H</sub> |
| subindex | name                 | type                        | price                | fetch            |
| 00       | Number of subindexes | unit8                       | 6                    | R O              |
| 01       | COB-ID               | unit32                      | 180 <sub>H</sub> +ID | R W              |
| 02       | TransmissionType     | unit8                       | 0                    | R W              |
| 03       | InhibitTime          | uint16                      | 0                    | R W              |
| 04       | CompatibilityEntry   | unit8                       | 0                    | R W              |
| 05       | E ventTimer          | uint16                      | 0                    | R W              |
| 06       | SYNCstartValue       | unit8                       | 0                    | R W              |

|                   |                                                                                                    |
|-------------------|----------------------------------------------------------------------------------------------------|
| Transmission Type | Transfer value is 0: data transmission changes and SYNC is received to return to the triggered PDO |
|                   | Transfer value is 1~240: the triggered PDO only after receiving the set value of                   |

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|                     |                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | S SYNC                                                                                                                                                                                                                                                                                                                                                    |
|                     | Transfer value 241~253: reserved (disabled)                                                                                                                                                                                                                                                                                                               |
|                     | Transmission value 254~255: trigger the PDO according to the timing period                                                                                                                                                                                                                                                                                |
| Inhibit Time        | Suphibit the forbidden time, that is, defines the minimum interval between two consecutive PDO transmission, to avoid the problem that the amount of data of high priority information always occupies the bus, and makes the other low priority data unable to compete with the bus. The forbidden time is defined by 16-bit unsigned integers in 100us. |
| Compatibility Entry | Reserve                                                                                                                                                                                                                                                                                                                                                   |
| Event Timer         | Event timing period, when the timing time is exceeded, a PDO transmission can be triggered (no trigger bit is required), in 1ms.                                                                                                                                                                                                                          |
| SYNC start Value    | Reserve                                                                                                                                                                                                                                                                                                                                                   |

### TPDO channel mapping

|          |                         |                        |       |       |
|----------|-------------------------|------------------------|-------|-------|
| Index    | 1A00 <sub>h</sub>       | TPDO 1 channel mapping |       |       |
|          | 1A01 <sub>h</sub>       | TPDO 2 channel mapping |       |       |
|          | 1A02 <sub>h</sub>       | TPDO 3 channel mapping |       |       |
|          | 1A03 <sub>h</sub>       | TPDO 4 channel mapping |       |       |
| Subindex | Name                    | Type                   | Price | Fetch |
| 00       | Number of subindexes    | unit8                  | 8     | RW    |
| 01       | PDO mapping parameter 1 | unit32                 | 0     | RW    |
| 02       | PDO mapping parameter 2 | unit32                 | 0     | RW    |
| 03       | PDO mapping parameter 3 | unit32                 | 0     | RW    |
| 04       | PDO mapping parameter 4 | unit32                 | 0     | RW    |
| 05       | PDO mapping parameter 5 | unit32                 | 0     | RW    |
| 06       | PDO mapping parameter 6 | unit32                 | 0     | RW    |
| 05       | PDO mapping parameter 7 | unit32                 | 0     | RW    |
| 06       | PDO mapping parameter 8 | unit32                 | 0     | RW    |

### RPDO Channel configuration

|                                                                     |       |          |        |
|---------------------------------------------------------------------|-------|----------|--------|
| The configuration parameters are the same as TPDO except for COB-ID |       |          |        |
| Name                                                                | Index | Subindex | COB-ID |

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|                              |                   |   |                      |
|------------------------------|-------------------|---|----------------------|
| RPDO1 channel configuration  | 1400 <sub>h</sub> | 1 | 200 <sub>H</sub> +ID |
| RPDO 2 channel configuration | 1401 <sub>h</sub> | 2 | 300 <sub>H</sub> +ID |
| RPDO 3 channel configuration | 1402 <sub>h</sub> | 3 | 400 <sub>H</sub> +ID |
| RPDO 4 channel configuration | 1403 <sub>h</sub> | 4 | 500 <sub>H</sub> +ID |

|                     |                                                                  |
|---------------------|------------------------------------------------------------------|
| Transmission Type   | Transfer value of 0: the RPDO function takes effect directly     |
|                     | Transfer value of 1~240: Retention of RPDO takes effect directly |
|                     | Transfer value 241~253: Retention of RPDO takes effect directly  |
|                     | Transfer value 254~255: Retention of RPDO takes effect directly  |
| Inhibit Time        | Reserve                                                          |
| Compatibility Entry | Reserve                                                          |
| Event Timer         | Reserve                                                          |
| SYNC start Value    | Reserve                                                          |

### RPDO Channel mapping

|       |                   |                        |
|-------|-------------------|------------------------|
| Index | 1600 <sub>h</sub> | RPDO 1 channel mapping |
|       | 1601 <sub>h</sub> | RPDO 2 channel mapping |
|       | 1602 <sub>h</sub> | RPDO 3 channel mapping |
|       | 1603 <sub>h</sub> | RPDO 4 channel mapping |

Heartbeat unit of 1ms, detailed explanation of the PDO parameters

| Transmission Type   | Transfer value                                                                                                                                                                                                                                                                                                                                            | Description                                                                | PDO       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------|
|                     | 0                                                                                                                                                                                                                                                                                                                                                         | Reserve                                                                    | —         |
|                     | 1~240                                                                                                                                                                                                                                                                                                                                                     | Synchronization mode: T SYNC is received and then sent to TPDO             | TPDO/RPDO |
|                     | 240~253                                                                                                                                                                                                                                                                                                                                                   | Reserve                                                                    | —         |
|                     | 254                                                                                                                                                                                                                                                                                                                                                       | Asynchronous mode: If the PDO content changes, trigger the PDO to be sent  | TPDO      |
|                     | 255                                                                                                                                                                                                                                                                                                                                                       | Asynchronous mode: automatically send the PDO according to the timing time | TPDO/RPDO |
| Inhibit Time        | Suphibit the forbidden time, that is, defines the minimum interval between two consecutive PDO transmission, to avoid the problem that the amount of data of high priority information always occupies the bus, and makes the other low priority data unable to compete with the bus. The forbidden time is defined by 16-bit unsigned integers in 100us. |                                                                            |           |
| Compatibility Entry | continue to have                                                                                                                                                                                                                                                                                                                                          |                                                                            |           |
| Event Timer         | Event timing period, after the timing time, a PDO transmission can be triggered (no trigger bit is required). The event timing period is defined by 16-bit                                                                                                                                                                                                |                                                                            |           |

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|                  |                           |
|------------------|---------------------------|
|                  | unsigned integers in 1ms. |
| SYNC start Value | Reserve                   |

PDO mapping process:

1. Set the content of the subindex 0 of the PDO corresponding to the mapping parameters (such as 1600 h or 1A00 h) as 0;
2. Modify the content of the subindex 1~4 of the PDO corresponding mapping parameters (such as 1600 h or 1A00 h), and map the data;
3. Set the content of the subindex 0 (such as 1600 h or 1A00 h) as a legitimate number (the number of objects mapped by the PDO).
4. the PDO mapping is completed.

The PDO can be delivered in several ways:

### 2.3 SYNC message

The synchronization object is used to control the synchronous transfer of data between network devices, and all nodes that support the synchronous PDO can receive (simultaneously) this message and use the object to synchronize with other nodes. So the SYNC messages are also called sync frames.

The general application mode is that the master node regularly sends the SYNC message, and performs the task synchronously after receiving it from the node. The COB-ID of the SYNC message was fixed at 80<sub>h</sub>, SYNC message can not transmit data to make the message all short, the general data length of 0, and the COB-ID can be read from 1005<sub>h</sub> of the object dictionary.

## Chapter Three Device Control

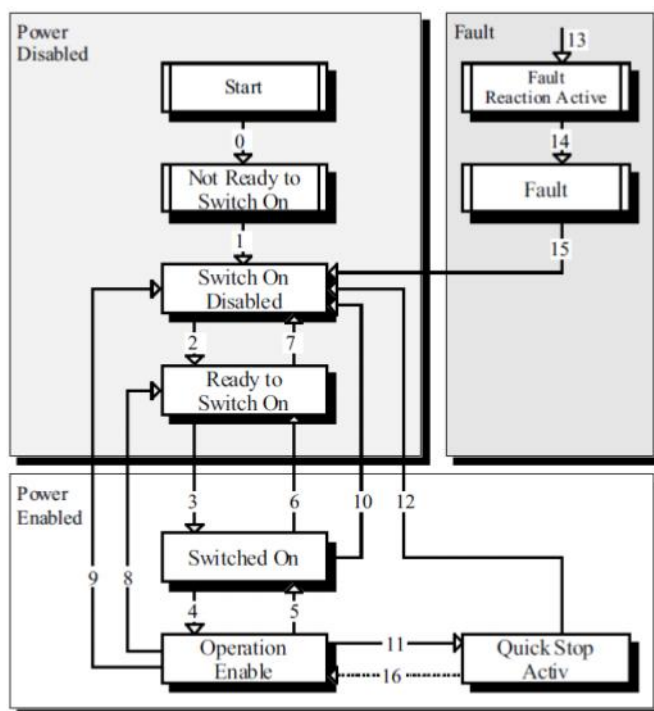
This section describes how the host controls the drive through the CANopen bus, such as controlling the servo ON or clearing the alarm.

The main station controls the drive by the control word (control word) and knows the current state of the drive by reading the status word (status word) of the drive. Some of the following terms are used in this section:

**State (state):** If the main power is active or an alarm occurs, the servo driver is in a different state. CANopen The status machine under the bus control will be highlighted in this section. For example: SWITCH \_ ON \_ DISABLED.

**State Transition (State Transfer):** The state machine also defines how to move from one state to another. The state transfer depends on the control word controlled by the master station or the drive itself, such as the drive alarm.

**Command (Command):** In order to start the State Transition (state transfer), we define the bit combinations of the control word, which are called the Command.



As shown in the figure above, the status machine can be divided into three parts: "Power Disabled" (main off), "Power Enabled" (main on) and "Fault". All states enter Fault after an alarm. After the drive is initialized, it enters the Switch on disabled state. In this state, CAN communication

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can be performed and the drive can be configured (for example, setting the working mode of the drive to "PP" mode). At this point, the main power supply is still closed, and the motor is not excited. After State Transition (state transmission) 2,3,4, enter the OPERATION ENABLE. At this time, the main power is on and the driver controls the motor according to the configured operating mode. Therefore, the state must be confirmed that the drive parameters and corresponding input values are properly configured as zero. State Transition (State transmission) 9 Complete the circuit main power. Once the drive alarm occurs, the state of the drive enters the FAULT.

| State name             | Illustration                                                                                                           |
|------------------------|------------------------------------------------------------------------------------------------------------------------|
| Not ready to switch on | The servo drive is initializing and cannot perform CAN communication.                                                  |
| Switch on disabled     | Servo drive initialization completes for CAN communication.                                                            |
| Ready to switch on     | The servo drive is waiting to enter Switch On State, the motor is not excited.                                         |
| Switched on            | Servo drive servo is ready, the main power is on.                                                                      |
| Operation enable       | The servo driver servo inputs the excitation signal to the motor and controls the motor according to the control mode. |
| Quick stop active      | The servo drive will stop in a set manner.                                                                             |
| Fault reaction active  | The servo driver detects an alarm and the motor still has excitation signal.                                           |
| Fault                  | The motor has no excitation signal.                                                                                    |

### Device control parameters

| Index             | Name                           | Type   | Attr |
|-------------------|--------------------------------|--------|------|
| 6040 <sub>h</sub> | Control word                   | uint16 | RW   |
| 6041 <sub>h</sub> | Status word                    | uint16 | RO   |
| 605A <sub>h</sub> | quick_stop_option_code         | int16  | RW   |
| 605B <sub>h</sub> | shutdown_option_code           | int16  | RW   |
| 605C <sub>h</sub> | disabled_operation_option_code | int16  | RW   |
| 605D <sub>h</sub> | halt_option_code               | int16  | RW   |
| 605E <sub>h</sub> | Fault_reaction_option_code     | int16  | RW   |

6040<sub>h</sub> : Control word

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| 15~11                 | 10~9     | 8    | 7           | 6~4                     | 3                | 2          | 1              | 0         |
|-----------------------|----------|------|-------------|-------------------------|------------------|------------|----------------|-----------|
| Manufacturer specific | Reserved | Halt | Fault reset | Operation mode specific | Enable operation | Quick stop | Enable voltage | Switch on |
| 0                     | 0        | 0    | 1           | 0                       | 1                | 1          | 1              | 1         |

(Support is 1, and will gradually support after version update)

Bit 0 to 3 and Bit 7

The transmission of the state machine starts from the corresponding control command consisting of these 5 bits.(The X means that the bit can be ignored.)

| Command           | The position of the control word |                  |            |                |           | Transitions |
|-------------------|----------------------------------|------------------|------------|----------------|-----------|-------------|
|                   | Fault reset                      | Enable operation | Quick stop | Enable voltage | Switch on |             |
| Shutdown          | 0                                | X                | 1          | 1              | 0         | 2,6,8       |
| Switch on         | 0                                | 0                | 1          | 1              | 1         | 3           |
| Switch on         | 0                                | 1                | 1          | 1              | 1         | 3           |
| Disable voltage   | 0                                | X                | X          | 0              | X         | 7,9,10,12   |
| Quick stop        | 0                                | X                | 0          | 1              | X         | 7,10,11     |
| Disable operation | 0                                | 0                | 1          | 1              | 1         | 5           |
| Enable operation  | 0                                | 1                | 1          | 1              | 1         | 4,16        |
| Fault reset       | 1                                | X                | X          | X              | X         | 15          |

Device control command table

Bit 4 ~ 6 and Bit 8

These 4 bits are defined differently in different mode control modes.

| B it | Control model          |                       |             |
|------|------------------------|-----------------------|-------------|
|      | Profile position mode  | Profile velocity mode | Homing mode |
| 4    | New_set_point          | /                     | /           |
| 5    | Change_set_immediately | /                     | /           |
| 6    | Abs /rel               | /                     | /           |
| 8    | /                      | /                     | /           |

6041<sub>h</sub> : Status word

| Position | Explain |
|----------|---------|
|----------|---------|

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|       |                         |
|-------|-------------------------|
| 0     | Ready to switch on      |
| 1     | Switched on             |
| 2     | Operation enabled       |
| 3     | Fault                   |
| 4     | Voltage enabled         |
| 5     | Quick stop              |
| 6     | Switch on disabled      |
| 7     | Warning                 |
| 8     | continue to have        |
| 9     | Remote                  |
| 10    | Target reached          |
| 11    | Internal limit active   |
| 13~12 | Operation mode specific |
| 15~14 | continue to have        |

Bit 0 ~ 3, Bit 5, and Bit 8

The combination of these bits indicates the state of the drive

| Value ( binary )    | State                  |
|---------------------|------------------------|
| xxxx xxxx x0xx 0000 | Not ready to switch on |
| xxxx xxxx x1xx 0000 | Switch on disabled     |
| xxxx xxxx x01x 0001 | Ready to switch on     |
| xxxx xxxx x01x 0011 | Switched on            |
| xxxx xxxx x01x 0111 | Operation enable       |
| xxxx xxxx x00x 0111 | Quick stop active      |
| xxxx xxxx x0xx 1111 | Fault reaction active  |
| xxxx xxxx x0xx 1000 | Fault                  |

Bit 4: When the bit is 1, the main power is on.

Bit 5: When this bit is 0, it means that the drive will stop according to the settings (605A h: quick\_stop\_option\_code).

Bit 7: When the bit is 1, the drive detects an alarm.

| ERROR _ CODE (16. x)    | Explain                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Index 603F <sub>h</sub> | <p>The corresponding bit is 0 indicates no error, and 1 indicates an error.</p> <p>The bit0 represents software errors such as motor run write FLASH etc.</p> <p>The bit1 represents the overvoltage.</p> <p>The bit2 represents the undervoltage.</p> <p>The bit4 represents a startup error.</p> <p>The bit5 represents a speed feedback error.</p> <p>The bit6 represents the overflow.</p> |



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|  |                                                                                                                                                                                                     |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>The bit16 represents an encoder communication error.</p> <p>The bit17 represents the too high motor temperature.</p> <p>The bit18 represents that the circuit board temperature is too high.</p> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Bit 9: When the bit is always set 1 in the enabling state, the Controlword of the drive can be processed.

Bit 10: This bit has different meanings in different control modes.

Profile Position Mode When the set position is reached, the position is set; when Halt starts to zero; when the new position is set, the position is cleared.

Profile Velocity Mode The bit is positioned when the speed reaches the given speed and when the Halt is activated to zero when the speed is reduced.

Bit11 : Internal limit active

When the bit is 1, it means that the internal torque exceeds the set value, or the position reaches the forward or reverse limit position. It can be determined by reading the object 60FDh (Digital Inputs).

Bit 12,13: The 2 bits have different meanings in different control modes

| Bit | Control model         |                       |             |
|-----|-----------------------|-----------------------|-------------|
|     | Profile position mode | Profile velocity mode | Homing mode |
| 12  | Set-point acknowledge | Speed                 | /           |
| 13  | Following error       | Max slippage error    | /           |

Other bits: all are reserved.

Mainly through several operation modes (operation mode) to achieve the control of various parameters of the servo motor, and to achieve some other auxiliary functions. Currently, the supported operating modes are:

|                                           |
|-------------------------------------------|
| Position Mode (Profile Position Mode)     |
| Speed Mode (Velocity Mode)                |
| Conile Speed Mode (Profile Velocity Mode) |
| Torque Pattern (Torque Mode)              |
| //                                        |

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|                                                            |
|------------------------------------------------------------|
| // Return to zero mode (Homing Mode).                      |
| Position Interpolation Mode (Interpolation Mode)           |
| Periodic position mode Cyclic synchronous position mode    |
| Cycle-speed mode (Cyclic synchronous speed mode            |
| // Cycle-based torque mode Cyclic synchronous CURRENT mode |

Control mode parameters

| Index             | Name                       | Type | Attr |
|-------------------|----------------------------|------|------|
| 6060 <sub>h</sub> | modes_of_operation         | int8 | RW   |
| 6061 <sub>h</sub> | modes_of_operation_display | int8 | RO   |

6060h : modes\_of\_operation

The control mode of the servo drive is determined by the mode \_ of \_ operation parameter.

| Value | Illustration                                            |      |
|-------|---------------------------------------------------------|------|
| 0     | Idle time (IDLE)                                        | None |
| 1     | Profile position mode (Profile Position Mode)           |      |
| 2     | Speed Mode (Velocity Mode)                              |      |
| 3     | Conile Speed Mode (Profile Velocity Mode)               |      |
| 4     | Torque Pattern (Profile Torque Mode)                    |      |
| 7     | Position Interpolation Mode (Interpolation Mode)        |      |
| 8     | Periodic position mode Cyclic synchronous position mode |      |
| 9     | Cycle-speed mode (Cyclic synchronous speed mode         |      |
| 10    | Periodic torque mode Cyclic synchronous CURRENT mode    |      |
| 11    | continue to have                                        |      |

6061h: modes\_of\_operation\_display

The current control mode of the servo drive can be known by reading the mode \_ of \_ operation \_ display parameter.

## Chapter Four Mode Control

### Illustration

Speed unit: 0.01 turn per second.

Acceleration: 0.01 rps / millisecond.

Location: cnt (The motor is converted to 65536cnt every time).

Contour velocity <maximum contour velocity <maximum motor speed.

Minimum position limit <target position <maximum position limit.

Zero-bit description setting:

Step 1: Read two indices (The value of the current location 0x6064 is A, and the value of the location offset 0x2008 is B).

Step 2: Add the two read values (current position value + offset position value =A + B).

Step 3: Set save (set the value of position offset 0x2008 to A + B, save).

Save parameter: index 0x2000, send 1 to save.

### Profile position mode (Profile Position Mode)

Dictionary objects involved:

| Index             | Subindex | Name                            | Data type | Visit |
|-------------------|----------|---------------------------------|-----------|-------|
| 603F <sub>h</sub> |          | error code                      | uint16    | RO    |
| 6040 <sub>h</sub> |          | control word                    | uint16    | RW    |
| 6041 <sub>h</sub> |          | status word                     | uint16    | RO    |
| 6060 <sub>h</sub> |          | control model                   | int8      | RW    |
| 6061 <sub>h</sub> |          | Control mode is shown           | int8      | RO    |
| 6062 <sub>h</sub> |          | User location requirement value | int32     | RO    |
| 6064 <sub>h</sub> |          | User location feedback          | int32     | RO    |
| 606C <sub>h</sub> |          | User actual speed feedback      | int32     | RO    |
| 607A <sub>h</sub> |          | target location                 | int32     | RW    |
| 607D <sub>h</sub> | 01       | Soft limit minimum position     | int32     | RW    |
| 607D <sub>h</sub> | 02       | Soft limit maximum position     | int32     | RW    |
| 6081 <sub>h</sub> |          | Outline speed                   | unit32    | RW    |
| 6083 <sub>h</sub> |          | Outline acceleration            | unit32    | RW    |
| 6084 <sub>h</sub> |          | The outline of the deceleration | unit32    | RW    |

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The contour position mode requires the following to the target index via PDO or SDO:

Step 1: start the motor

| Index                                                                                          | Step                                                      | Set value |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------|
| 6060                                                                                           | The switching operation mode is the contour position mode | 1         |
| 6040                                                                                           | Enable motor                                              | 6->15     |
| Complete the above steps to enter the contour position mode and maintain the current position. |                                                           |           |

Step 2: Set the parameters

| Index                                                                                                                                                                                                                                                                                                       | Step                                                              | Set value |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------|
| 607D <sub>h</sub>                                                                                                                                                                                                                                                                                           | Set the soft limit min. position and the soft limit max. position | ±65536000 |
| 6081 <sub>h</sub>                                                                                                                                                                                                                                                                                           | Outline speed                                                     | 2000      |
| 6083 <sub>h</sub>                                                                                                                                                                                                                                                                                           | Outline acceleration                                              | 10        |
| 6084 <sub>h</sub>                                                                                                                                                                                                                                                                                           | The outline of the deceleration                                   | 10        |
| The above parameters will be updated to the internal path planning and recalculated as the internal unit after the motor updates the target position. The motor will plan the trapezoidal path of the motor from the current position to the target position at the set acceleration and deceleration speed |                                                                   |           |

Step 3: Set the target position

| Index                                                                                                                                                                                            | Step                                                                                                                                                                                            | Set value |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 607A <sub>h</sub>                                                                                                                                                                                | target location                                                                                                                                                                                 | 1000000   |
| 6040 <sub>h</sub>                                                                                                                                                                                | According to the usage of the control word, choose absolute position control, relative position control, and update the position immediately, and update the position after completing the path | /         |
| After updating the position, the set parameter motor will plan the trapezoidal path of the motor from the current position to the target position at the set acceleration and deceleration speed |                                                                                                                                                                                                 |           |

Status read:

| Index             | Name                       | Meaning |
|-------------------|----------------------------|---------|
| 603F <sub>h</sub> | Error code                 |         |
| 6041 <sub>h</sub> | Status word                |         |
| 6061 <sub>h</sub> | Control mode is shown      |         |
| 6064 <sub>h</sub> | User location feedback     |         |
| 606C <sub>h</sub> | User actual speed feedback |         |

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## Speed control (Velocity Mode)

Dictionary objects involved:

| Index             | Subindex | Name                            | Data type | Visit |
|-------------------|----------|---------------------------------|-----------|-------|
| 603F <sub>h</sub> |          | Error code                      | uint16    | RO    |
| 6040 <sub>h</sub> |          | Control word                    | uint16    | RW    |
| 6041 <sub>h</sub> |          | Status word                     | uint16    | RO    |
| 6060 <sub>h</sub> |          | Control model                   | int8      | RW    |
| 6061 <sub>h</sub> |          | Control mode is shown           | int8      | RO    |
| 6064 <sub>h</sub> |          | User location feedback          | int32     | RO    |
| 606B <sub>h</sub> |          | User speed instruction value    | int32     | RO    |
| 606C <sub>h</sub> |          | User actual speed feedback      | int32     | RO    |
| 606D <sub>h</sub> |          | The speed reaches the threshold | uint16    | RW    |
| 606E <sub>h</sub> |          | Speed arrival time              | uint16    | RW    |
| 606F <sub>h</sub> |          | Zero speed threshold            | uint16    | RW    |
| 6070 <sub>h</sub> |          | Zero-speed threshold time       | uint16    | RW    |
| 6083 <sub>h</sub> |          | Outline acceleration            | uint32    | RW    |
| 6044 <sub>h</sub> |          | The outline of the deceleration | uint32    | RW    |
| 607F              |          | Maximum contour speed           |           |       |
| 60FF <sub>h</sub> |          | Target speed                    | int32     | R W   |

Speed use a tutorial using PDO control or SDO control

Step 1: start the motor

| Index                                                                               | Step                                       | Set value |
|-------------------------------------------------------------------------------------|--------------------------------------------|-----------|
| 6060                                                                                | Switching operation mode to the speed mode | 2         |
| 6040                                                                                | Enable motor                               | 6->15->31 |
| Complete the above steps to enter the contour speed mode and maintain a speed of 0. |                                            |           |

Step 2: Set the parameters

| Index                                                                                              | Step                            | Set value |
|----------------------------------------------------------------------------------------------------|---------------------------------|-----------|
| 6081 <sub>h</sub>                                                                                  | Maximum contour speed           | 2000      |
| 6083 <sub>h</sub>                                                                                  | Outline acceleration            | 10        |
| 6084 <sub>h</sub>                                                                                  | The outline of the deceleration | 10        |
| 60FF <sub>h</sub>                                                                                  | Target speed                    | 1000      |
| The target speed and the new rear motor will reach the target speed in a trapezoidal acceleration. |                                 |           |

Status read:

| Index             | Name       | Meaning |
|-------------------|------------|---------|
| 603F <sub>h</sub> | Error code |         |

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|                   |                            |  |
|-------------------|----------------------------|--|
| 6041 <sub>h</sub> | Status word                |  |
| 6061 <sub>h</sub> | Control mode is shown      |  |
| 6064 <sub>h</sub> | User location feedback     |  |
| 606C <sub>h</sub> | User actual speed feedback |  |

### Conile Speed Control (ProfileVelocity Mode)

Dictionary objects involved:

| Index             | Subindex | Name                                   | Data type | Visit |
|-------------------|----------|----------------------------------------|-----------|-------|
| 603F <sub>h</sub> |          | error code                             | uint16    | RO    |
| 6040 <sub>h</sub> |          | control word                           | uint16    | RW    |
| 6041 <sub>h</sub> |          | status word                            | uint16    | RO    |
| 6060 <sub>h</sub> |          | control model                          | int8      | RW    |
| 6061 <sub>h</sub> |          | Control mode is shown                  | int8      | RO    |
| 6063 <sub>h</sub> |          | Motor position feedback                | int32     | RO    |
| 6064 <sub>h</sub> |          | User location feedback                 | int32     | RO    |
| 6065 <sub>h</sub> |          | Maximum deviation of the user location | unit32    | RW    |
| 606B <sub>h</sub> |          | User speed instruction value           | int32     | RO    |
| 606C <sub>h</sub> |          | User actual speed feedback             | int32     | RO    |
| 606D <sub>h</sub> |          | The speed reaches the threshold        | unit16    | RW    |
| 606E <sub>h</sub> |          | Speed arrival time                     | unit16    | RW    |
| 606F <sub>h</sub> |          | Zero speed threshold                   | unit16    | RW    |
| 6070 <sub>h</sub> |          | Zero-speed threshold time              | unit16    | RW    |
| 607D <sub>h</sub> | 01       | Soft limit minimum position            | int32     | RW    |
| 607D <sub>h</sub> | 02       | Soft limit maximum position            | int32     | RW    |
| 6083 <sub>h</sub> |          | Outline acceleration                   | unit32    | RW    |
| 6044 <sub>h</sub> |          | The outline of the deceleration        | uint32    | RW    |
| 6094 <sub>h</sub> | 01       | Speed factor: Molecules                | unit32    | RW    |
| 6094 <sub>h</sub> | 02       | Speed factor: the feed constant        | unit32    | RW    |
| 60FF <sub>h</sub> |          | Target speed                           | int32     | RW    |

Currently no difference between mode 3 and mode 2.

Outline speed using tutorial, using PDO control or SDO control

Step 1: start the motor

| Index                                                                                 | Step                                                   | Set value |
|---------------------------------------------------------------------------------------|--------------------------------------------------------|-----------|
| 6060                                                                                  | The switching operation mode is the contour speed mode | 3         |
| 6040                                                                                  | Enable motor                                           | 6->15->31 |
| Complete the above steps to enter the contour speed mode and maintain the speed of 0. |                                                        |           |

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## Step 2: Set the parameters

| Index                                                                                                                        | Step                            | Set value |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------|
| 6081 <sub>h</sub>                                                                                                            | Maximum contour speed           | 2000      |
| 6083 <sub>h</sub>                                                                                                            | Outline acceleration            | 10        |
| 6084 <sub>h</sub>                                                                                                            | The outline of the deceleration | 10        |
| 60FF <sub>h</sub>                                                                                                            | Target speed                    | 1000      |
| After the target speed is updated, the electrical opportunity reaches the target speed in a trapezoidal acceleration manner. |                                 |           |

## Status read:

| Index             | Name                       | Meaning |
|-------------------|----------------------------|---------|
| 603F <sub>h</sub> | Error code                 |         |
| 6041 <sub>h</sub> | Status word                |         |
| 6061 <sub>h</sub> | Control mode is shown      |         |
| 6064 <sub>h</sub> | User location feedback     |         |
| 606C <sub>h</sub> | User actual speed feedback |         |

## Profile torque control (Profile Torque Mode)

Current torque is not calibrated, using the current calculation unit: milliampere

Dictionary objects involved:

| Index             | Subindex | Name                       | Data type | Visit |
|-------------------|----------|----------------------------|-----------|-------|
| 603F <sub>h</sub> |          | error code                 | uint16    | RO    |
| 6040 <sub>h</sub> |          | control word               | uint16    | RW    |
| 6041 <sub>h</sub> |          | status word                | uint16    | RO    |
| 6060 <sub>h</sub> |          | control model              | int8      | RW    |
| 6061 <sub>h</sub> |          | Control mode is shown      | int8      | RO    |
| 6064 <sub>h</sub> |          | User location feedback     | int32     | RO    |
| 606C <sub>h</sub> |          | User actual speed feedback | int32     | RO    |
| 6071 <sub>h</sub> |          | Target torque              | int16     | RW    |
| 6072 <sub>h</sub> |          | acrot orque                | int16     | RW    |
| 6074 <sub>h</sub> |          | Demand torque              | int16     | RO    |
| 6087 <sub>h</sub> |          | Torque slope               | uint32    | RW    |

## Step 1: start the motor

| Index                                             | Step                                        | Set value |
|---------------------------------------------------|---------------------------------------------|-----------|
| 6060                                              | Switch operation mode to the contour torque | 4         |
| 6040                                              | Enable motor                                | 6->15->31 |
| Complete the above steps to enter contour torque. |                                             |           |

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Step 2: Set the parameters

| Index                                                                                                                            | Step                                                             | Set value |
|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------|
| 6072 <sub>h</sub>                                                                                                                | Maximum torque (power default is 0)                              | 1000      |
| 6071 <sub>h</sub>                                                                                                                | Target torque (current value mA)                                 | 1000      |
| 6074                                                                                                                             | Output target torque                                             |           |
| 6087                                                                                                                             | Torque slope (slope of change of output torque to target torque) | 1         |
|                                                                                                                                  |                                                                  |           |
| After the target torque is updated, the electrical opportunity will reach the target torque in a trapezoidal acceleration manner |                                                                  |           |

The synchronous control mode 8,9, and 10 adopts the same parameters and 1,3, and 4.

### **Periodic position mode (Cyclic synchronous position mode)**

The index used is the same as the contour position (difference: no contour acceleration, contour deceleration).

### **Periodic speed mode (Cyclic synchronous speed mode)**

The index used is consistent with the speed mode.

### **Periodic torque mode (Cyclic synchronous CURRENT mode)**

The index used is the same with contour torque.



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## Chapter Five Index table

| User-defined                       |       |          |           |                          |
|------------------------------------|-------|----------|-----------|--------------------------|
|                                    | Index | Subindex | Data type | Reading and writing type |
| User save                          | 2000  | 00       | UNS32     | RW                       |
| User Settings                      | 2001  | 00       | UNS32     | RW                       |
| Factory setting                    | 2002  | 00       | UNS32     | RW                       |
| Plate temperature protection value | 2003  | 00       | INTEGER16 | RW                       |
| Motor temperature protection value | 2004  | 00       | INTEGER16 | RW                       |
| Parameter 1                        | 2005  | 00       | UNS32     | RW                       |
| Motor ID                           | 2006  | 00       | UNS16     | RW                       |
| Baud rate                          | 2007  | 00       | UNS16     | RW                       |
| Position offset                    | 2008  | 00       | INTEGER32 | RW                       |
| Maximum current                    | 2009  | 00       | INTEGER32 | RW                       |
| Minimum current                    | 200A  | 00       | INTEGER32 | RW                       |
| Current absolute value             | 200B  | 00       | INTEGER32 | RO                       |
| The current temperature            | 200C  | 00       | INTEGER16 | RO                       |
| Current plate temperature          | 200D  | 00       | INTEGER16 | RO                       |
| Current voltage                    | 200E  | 00       | UNS16     | RO                       |
| Communication conversion           | 20FE  | 00       | INTEGER32 | RW// test                |
| Program version                    | 20FF  | 00       | INTEGER32 | RO                       |

| CIA 402         |       |          |           |                 |
|-----------------|-------|----------|-----------|-----------------|
|                 | Index | Subindex | Data type | Windows default |
| Error code      | 603F  | 00       | UNS32     | RO              |
| Control word    | 6040  | 00       | UNS16     | RW              |
| Status word     | 6041  | 00       | UNS16     | RO              |
| Operator schema | 6060  | 00       | INTEGER8  | RW              |
| Mode display    | 6061  | 00       | INTEGER8  | RO              |
|                 |       |          |           |                 |
| Actual location | 6064  | 00       | INTEGER32 | RO              |
| Actual velocity | 606C  | 00       | INTEGER32 | RO              |
| Actual current  | 6078  | 00       | INTEGER16 | RO              |
| Target torque   | 6071  | 00       | INTEGER16 | RW              |
| Max. torque     | 6072  | 00       | INTEGER16 | RW              |

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|                                 |      |    |           |    |
|---------------------------------|------|----|-----------|----|
| Target torque                   | 6074 | 00 | INTEGER16 | RO |
| Target location                 | 607A | 00 | INTEGER32 | RW |
| Software location limit         | 607D | 01 | INTEGER32 | RW |
|                                 |      | 02 | INTEGER32 | RW |
| Maximum speed                   | 607F | 00 | UNS32     | RW |
| Outline speed                   | 6081 | 00 | UNS32     | RW |
| Outline acceleration            | 6083 | 00 | UNS32     | RW |
| The outline of the deceleration | 6084 | 00 | UNS32     | RW |
| Torque slope                    | 6074 | 00 | UNS32     | RW |
| Electric current loop PID       | 60F6 | 01 | INTEGER16 | RO |
|                                 |      | 02 | INTEGER16 | RO |
| Speed ring PID                  | 60F9 | 01 | INTEGER16 | RW |
|                                 |      | 02 | INTEGER16 | RW |
| Position ring PID               | 60FB | 01 | INTEGER16 | RW |
|                                 |      | 02 | INTEGER16 | RW |
|                                 |      | 03 | INTEGER16 | RW |
| Target speed                    | 60FF | 00 | INTEGER32 | RW |

6 motor full indexes(Some index motor control is not used, marked as: none)

### Index : 0x2000 User save

|                                                                                                                                 |                     |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Subindex: 00                                                                                                                    | Data type: uint32_t |
| Save the parameters of the motor index table, issue 1 to save the parameters, and automatically clear the zero after completion |                     |
| Default: 00                                                                                                                     |                     |

### Index: 0x2001 User read

|                                                                                                                                 |                     |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Subindex: 00                                                                                                                    | Data type: uint32_t |
| Read the parameters of the motor index table, issue the 1 reading parameters, and automatically clear the zero after completion |                     |
| Default: 00                                                                                                                     |                     |

### Index: 0x2002 Factory setting

|                                                                                                              |                     |
|--------------------------------------------------------------------------------------------------------------|---------------------|
| Subindex: 00                                                                                                 | Data type: uint32_t |
| Restore the factory Settings, issue the 1 read parameters, and automatically clear the zero after completion |                     |
| Default: 00                                                                                                  |                     |

### Index: 0x2003 Plate temperature protection value

|                                                                                                                                                 |                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Subindex: 00                                                                                                                                    | Data type: int16_t |
| Circuit board temperature protection alarm value, when the temperature sensor is above this value, the motor stops, per unit of degrees Celsius |                    |
| Default: 80                                                                                                                                     |                    |

### Index: 0x2004 Motor temperature protection value

|              |                    |
|--------------|--------------------|
| Subindex: 00 | Data type: int16_t |
|--------------|--------------------|

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|                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------|
| Motor coil temperature protection value, when the temperature sensor is above this value, the motor stops, unit degrees Celsius |
| Default value: 110                                                                                                              |

## Index: 0x2005 Other parameters (none)

|               |                   |
|---------------|-------------------|
| Subindex: 00  | Data type: uint32 |
| Not available |                   |
| Default: 0    |                   |

## Index: 0x2006 any power-generating or power-driven machine ID

|                                                     |                   |
|-----------------------------------------------------|-------------------|
| Subindex: 00                                        | Data type: uint16 |
| Motor ID, (1~127) modified, only after preservation |                   |
| Default: 1                                          |                   |

## Index: 0x2007 Porter rate setting

|                                                                                                                                |                   |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Subindex: 00                                                                                                                   | Data type: uint16 |
| Use to set the communication rate of the motor, optional value: 1000,500,250,125,100,50, modified, only effective after saving |                   |
| Default value: 1000                                                                                                            |                   |

## Index: 0x2008 Location offset

|                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Subindex: 00                                                                                                                                                                                                                                                                                                                                                                                                       | Data type: int32 |
| Function: Return to zero at the set read position, set the zero and start position, and it takes effect directly after sending.<br>Formula: read motor position = encoder actual position-position offset.<br>Sets the offset required for zero = the read offset+ the read motor position.<br>Method: the current position of the read 0x6040 is A, and the offset of the read 0x2008 is B. The sent data is A+B. |                  |
| Default: 0                                                                                                                                                                                                                                                                                                                                                                                                         |                  |

## Index: 0x2009 Maximum current setting

|                                                                                              |                  |
|----------------------------------------------------------------------------------------------|------------------|
| Subindex: 00                                                                                 | Data type: int32 |
| The motor running maximum current, absolute value less than the value of index 0x 200B in mA |                  |
| Default value: by model number                                                               |                  |

## Index: 0x200A Minimum current setting

|                                                                                           |                  |
|-------------------------------------------------------------------------------------------|------------------|
| Subindex: 00                                                                              | Data type: int32 |
| Motor running minimum current, absolute value less than the value of index 0x 200B, in mA |                  |
| Default value: by model number                                                            |                  |

## Index: 0x200B Current absolute value

|                                                                                       |                  |
|---------------------------------------------------------------------------------------|------------------|
| Subindex: 00                                                                          | Data type: int32 |
| The reference current value allowed by the motor (according to the motor model) in mA |                  |
| Default value: by model number                                                        |                  |

## Index: 0x200C Coil temperature value

|              |                  |
|--------------|------------------|
| Subindex: 00 | Data type: int16 |
|--------------|------------------|

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|                                                         |
|---------------------------------------------------------|
| Current motor coil temperature value in degrees Celsius |
| Default: 0                                              |

## Index:0x200D **Circuit board temperature value**

|                                                             |                  |
|-------------------------------------------------------------|------------------|
| Subindex: 00                                                | Data type: int16 |
| Current circuit board temperature value, in degrees Celsius |                  |
| Default: 0                                                  |                  |

## Index:0x200E **Motor bus voltage**

|                                            |                   |
|--------------------------------------------|-------------------|
| Subindex: 00                               | Data type: unit16 |
| Motor bus voltage, which is used in unit V |                   |
| Default: 36                                |                   |

## Index:0x20FF **Motor CANOPEN program updated version**

|                                          |                  |
|------------------------------------------|------------------|
| Subindex: 00                             | Data type: int32 |
| Program update version time              |                  |
| Default value: Based on the version date |                  |

## Index:0x603F **error condition**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Subindex: 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Data type: unit16 |
| <p>A digit of 0 indicates no error, and a digit of 1 indicates an error.<br/> The bit0 represents software errors such as motor run write FLASH etc.<br/> The bit1 represents the overvoltage.<br/> The bit2 represents the undervoltage<br/> The bit4 represents a startup error.<br/> The bit5 represents a speed feedback error.<br/> The bit6 represents the overflow.<br/> The bit16 represents an encoder communication error.<br/> The bit17 represents the too high motor temperature.<br/> The bit18 represents that the circuit board temperature is too high.</p> |                   |
| Default: 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |

## Index:0x6040 **control word**

|                                                 |                   |
|-------------------------------------------------|-------------------|
| Subindex: 00                                    | Data type: unit16 |
| Control motor enable, disability, remove errors |                   |
|                                                 |                   |

## Index:0x6041 **status word**

|                           |                   |
|---------------------------|-------------------|
| Subindex: 00              | Data type: unit16 |
| Displays the motor status |                   |
| Default: 0                |                   |

## Index:0x605A **Fast shutdown (none)**

|                                                     |                  |
|-----------------------------------------------------|------------------|
| Subindex: 00                                        | Data type: int16 |
| Different values represent different shutdown modes |                  |
| Default: 80                                         |                  |

## Index:0x605B **(not have)**

|                                                     |                  |
|-----------------------------------------------------|------------------|
| Subindex: 00                                        | Data type: int16 |
| Different values represent different shutdown modes |                  |

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|                     |                                                              |
|---------------------|--------------------------------------------------------------|
|                     | Default: 80                                                  |
| <b>Index:0x605C</b> | <b>(not have)</b>                                            |
|                     | Subindex: 00      Data type: int16                           |
|                     | Different values represent different shutdown modes          |
|                     | Default: 80                                                  |
| <b>Index:0x605D</b> | <b>Pause way (not have)</b>                                  |
|                     | Subindex: 00      Data type: int16                           |
|                     | Different values represent different shutdown modes          |
|                     | Default: 80                                                  |
| <b>Index:0x605E</b> | <b>(not have)</b>                                            |
|                     | Subindex: 00      Data type: int16                           |
|                     | Different values represent different shutdown modes          |
|                     | Default: 0                                                   |
| <b>Index:0x6060</b> | <b>operator schema</b>                                       |
|                     | Subindex: 00      Data type: int8                            |
|                     | The motion mode to be set by the motor                       |
|                     | Default: 8                                                   |
| <b>Index:0x6061</b> | <b>Display mode</b>                                          |
|                     | Subindex: 00      Data type: int8                            |
|                     | Show the motion pattern of the motor                         |
|                     | Default: 0                                                   |
| <b>Index:0x6062</b> | <b>User location instructions (not have)</b>                 |
|                     | Subindex: 00      Data type: int32                           |
|                     | User demand value                                            |
|                     | Default: 0                                                   |
| <b>Index:0x6063</b> | <b>Motor position feedback (none)</b>                        |
|                     | Subindex: 00      Data type: int32                           |
|                     | Feedback to the motor encoder position                       |
|                     | Default: 0                                                   |
| <b>Index:0x6064</b> | <b>User location feedback</b>                                |
|                     | Subindex: 00      Data type: int32                           |
|                     | User location feedback                                       |
|                     | Default: 0                                                   |
| <b>Index:0x6065</b> | <b>Excessive threshold for deviation of user position</b>    |
|                     | Subindex: 00      Data type: int32                           |
|                     | Whether the position deviation value is within the set range |
|                     | Default: 0                                                   |
| <b>Index:0x6066</b> | <b>Position deviation time window</b>                        |
|                     | Subindex: 00      Data type: ui nt16                         |
|                     | unit ms                                                      |
|                     | Default: 0                                                   |

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## Index:0x6067 Position reaches the threshold value

|              |           |
|--------------|-----------|
| Subindex: 00 | data type |
|              |           |
| Default: 8   |           |

## Index:0x6068 Location arrival time

|              |           |
|--------------|-----------|
| Subindex: 00 | data type |
|              |           |
| Default: 8   |           |

## Index:0x6069 Actual value of the speed sensor (not have)

|                                  |           |
|----------------------------------|-----------|
| Subindex: 00                     | data type |
| Actual value of the speed sensor |           |
| Default: 8                       |           |

## Index:0x606B User speed command (none)

|                              |           |
|------------------------------|-----------|
| Subindex: 00                 | data type |
| User speed instruction value |           |
| Default: 8                   |           |

## Index:0x606C User actual speed feedback

|                            |           |
|----------------------------|-----------|
| Subindex: 00               | data type |
| User actual speed feedback |           |
| Default: 8                 |           |

## Index:0x606D The speed reaches the threshold

|                |           |
|----------------|-----------|
| Subindex: 00   | data type |
| Speed of motor |           |
| Default: 0     |           |

## Index:0x606E Speed reaches the threshold time window

|                |           |
|----------------|-----------|
| Subindex: 00   | data type |
| Speed of motor |           |
| Default: 0     |           |

## Index:0x606F Zero speed threshold

|                |           |
|----------------|-----------|
| Subindex: 00   | data type |
| Speed of motor |           |
| Default: 8     |           |

## Index:0x6070 Zero speed threshold time

|              |           |
|--------------|-----------|
| Subindex: 00 | data type |
|              |           |
| Default: 0   |           |

## Index:0x6071 Target torque

|              |           |
|--------------|-----------|
| Subindex: 00 | data type |
|              |           |
| Default: 0   |           |

## Index:0x6072 maximum torque

|                             |           |
|-----------------------------|-----------|
| Subindex: 00                | data type |
| Limits of the target torque |           |

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|                     |                                                                         |
|---------------------|-------------------------------------------------------------------------|
|                     | Default: 8                                                              |
| <b>Index:0x6074</b> | <b>The user gives a torque</b>                                          |
|                     | Subindex: 00 data type                                                  |
|                     | Default: 0                                                              |
| <b>Index:0x6077</b> | <b>Time torque feedback</b>                                             |
|                     | Subindex: 00 data type                                                  |
|                     | Default: 0                                                              |
| <b>Index:0x6078</b> | <b>The actual current</b>                                               |
|                     | Subindex: 00 data type                                                  |
|                     | UVW coil current, unit in MA, missing torque current conversion formula |
|                     | Default: 8                                                              |
| <b>Index:0x607A</b> | <b>target location</b>                                                  |
|                     | Subindex: 00 data type                                                  |
|                     | In position mode, to set the arrival location                           |
|                     | Default: 8                                                              |
| <b>Index:0x607C</b> | <b>Origin offset (none)</b>                                             |
|                     | Subindex: 00 data type                                                  |
|                     | In position mode, to set the arrival location                           |
|                     | Default: 8                                                              |
| <b>Index:0x607D</b> | <b>Software location limit</b>                                          |
|                     | Subindex: 01 data type                                                  |
|                     | Minimum lower limit of motor position: $65536 * 101 * 100$              |
|                     | Default: 8                                                              |
|                     | Subindex: 02 data type                                                  |
|                     | Maximum limit of motor position: $65536 * 101 * 100$                    |
|                     | Default: 8                                                              |
| <b>Index:0x607F</b> | <b>Maximum contour speed</b>                                            |
|                     | Subindex: 00 data type                                                  |
|                     | The highest rotational speed of the motor                               |
|                     | Default value: 5,000                                                    |
| <b>Index:0x6080</b> | <b>Maximum speed (None)</b>                                             |
|                     | Subindex: 00 data type                                                  |
|                     | The highest rotational speed of the motor                               |
|                     | Default value: 5,000                                                    |
| <b>Index:0x6081</b> | <b>Outline speed</b>                                                    |
|                     | Subindex: 00 data type                                                  |
|                     | Speed in contour mode and less than maximum speed                       |

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|                     |
|---------------------|
| Default value: 2000 |
|---------------------|

## Index:0x6082 (not have)

|                                                   |           |
|---------------------------------------------------|-----------|
| Subindex: 00                                      | data type |
| Speed in contour mode and less than maximum speed |           |
| Default: 0                                        |           |

## Index:0x6083 Outline acceleration

|                                                                                                                       |           |
|-----------------------------------------------------------------------------------------------------------------------|-----------|
| Subindex: 00                                                                                                          | data type |
| The current speed in the speed mode increases per millisecond from the target speed, and the position mode is useless |           |
| Default: 1                                                                                                            |           |

## Index:0x6084 The outline of the deceleration

|                                                                                                                       |           |
|-----------------------------------------------------------------------------------------------------------------------|-----------|
| Subindex: 00                                                                                                          | data type |
| The current speed in the speed mode decreases from the target speed per millisecond, and the position mode is useless |           |
| Default: 1                                                                                                            |           |

## Index:0x6085 Quick to stop the deceleration (not have)

|                                |           |
|--------------------------------|-----------|
| Subindex: 00                   | data type |
| Quick to stop the deceleration |           |
| Default: 1                     |           |

## Index:0x6086 (not have)

|              |           |
|--------------|-----------|
| Subindex: 00 | data type |
|              |           |
| Default: 1   |           |

## Index:0x6087 Torque slope coefficient

|                                                                           |           |
|---------------------------------------------------------------------------|-----------|
| Subindex: 00                                                              | data type |
| Target torque to output torque increases the output slope per millisecond |           |
| Default: 1                                                                |           |

## Index:0x6093 Location Factor (None)

|              |           |
|--------------|-----------|
| Subindex: 00 | data type |
|              |           |
| Default: 1   |           |

## Index:0x6094 Speed Factor (None)

|              |           |
|--------------|-----------|
| Subindex: 00 | data type |
|--------------|-----------|



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|            |
|------------|
|            |
| Default: 1 |

## Index:0x6097      Contour acceleration factor (none)

|              |           |
|--------------|-----------|
| Subindex: 00 | data type |
|              |           |
| Default: 1   |           |

## Index:0x6098      Return to zero mode (none)

|              |           |
|--------------|-----------|
| Subindex: 00 | data type |
|              |           |
| Default: 1   |           |

## Index:0x60F6      electric current loop PID (read only)

Subindex: 01      Windows default:

Function: the proportion of the current ring P.

Formula: /

Method: read only

|                                      |           |
|--------------------------------------|-----------|
| Subindex: 00                         | data type |
| The proportion of the current ring P |           |
| Default: 1                           |           |

|                                   |           |
|-----------------------------------|-----------|
| Subindex: 00                      | data type |
| Integration I of the current ring |           |
| Default: 1                        |           |

|                                    |           |
|------------------------------------|-----------|
| Subindex: 00                       | data type |
| Differential D of the current ring |           |
| Default: 1                         |           |

## Index:0x60F9      speed ring PID

|                                        |           |
|----------------------------------------|-----------|
| Subindex: 01                           | data type |
| The proportion of the velocity ring, P |           |
| Default: 1                             |           |

|                                    |           |
|------------------------------------|-----------|
| Subindex: 02                       | data type |
| Integration I of the velocity loop |           |
| Default: 1                         |           |

## Index:0x60FB      Position ring PID

|                                    |           |
|------------------------------------|-----------|
| Subindex: 02                       | data type |
| Integration I of the position ring |           |

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Default: 1

|                                    |           |
|------------------------------------|-----------|
| Subindex: 02                       | data type |
| Integration I of the position ring |           |
| Default: 1                         |           |

|                                    |           |
|------------------------------------|-----------|
| Subindex: 02                       | data type |
| Integration I of the position ring |           |
| Default: 1                         |           |

## Index:0x60FC Speed demand value

|                                    |           |
|------------------------------------|-----------|
| Subindex: 02                       | data type |
| Integration I of the position ring |           |
| Default: 1                         |           |

## Index:0x60FF Target speed

Subindex: 00

Default: 0

Function: Set the target speed in the motor speed mode.

Formula: Target speed = (output target speed (degree per second) \* deceleration ratio \* 100) / 360;

Target speed = (output target speed (RPM) \* deceleration ratio \* 100) / (360 \* 6);

Method: enable the motor in speed mode.

## Chapter Six Control case

### SDO read and write parameters

Parameters to override: control word 0x6040, Controlword, data type UNS 16

a slight pause in reading

| Frame ID | DLC | Data 0 | Data 1 | Data 2 | Data 3 | Data 4 | Data 5 | Data 6 | Data 7 |
|----------|-----|--------|--------|--------|--------|--------|--------|--------|--------|
| 0x60X    | 8   | 40     | 40     | 60     | 00     | 00     | 00     | 00     | 00     |
| 0x 58X   | 8   | 4B     | 40     | 60     | 00     | 00     | 00     | 00     | 00     |

write

| frame ID | DLC | Data 0 | Data 1 | Data 2 | Data 3 | Data 4 | Data 5 | Data 6 | Data 7 |
|----------|-----|--------|--------|--------|--------|--------|--------|--------|--------|
| 0x 60X   | 8   | 2B     | 40     | 60     | 00     | 00     | 00     | 00     | 00     |
| 0x 58X   | 8   | 60     | 40     | 60     | 00     | XX     | XX     | 00     | 00     |

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## PDO read and write format

| frame ID | DLC | Data 0 | Data 1 | Data 2 | Data 3 | Data 4 | Data 5 | Data 7 | Data 8 |
|----------|-----|--------|--------|--------|--------|--------|--------|--------|--------|
| ID       | DL  | data   | data   | data   | data   | data   | data   | data   | data   |

frame ID:

PDO channel 1: the frame ID for sending TPDO \_ 1 data = 0x 200 + \$N ODEID  
 Frame ID that receives the RPDO \_ 1 data = 0x 180 + \$N ODEID  
 PDO channel 2: the frame ID for sending the TPDO \_ 2 data = 0x 300 + \$N ODEID  
 Frame ID that receives the RPDO \_ 2 data = 0x 280 + \$N ODEID  
 PDO channel 3: the frame ID for sending the TPDO \_ 3 data = 0x 400 + \$N ODEID  
 Frame ID that receives the RPDO \_ 3 data = 0x 380 + \$N ODEID  
 PDO channel 4: the frame ID for sending TPDO \_ 4 data = 0x 500 + \$N ODEID  
 Frame ID that receives the RPDO \_ 4 data = 0x 480 + \$N ODEID

D LC:

The DLC of the PDO is the sum of the N data lengths sent

Data: directly distribute the data values according to the address of the pre-configured index

PID revise

Read speed loop P

43 F9 60 01 00 00 00 00

Read speed loop I

43 F9 60 02 00 00 00 00

Write speed loop P

2B F9 60 01 E8 03 00 00

Write speed loop I

2B F9 60 02 E8 03 00 00

Read position loop P

43 FB 60 01 00 00 00 00

Read position ring D

43 FB 60 01 00 00 00 00

Write position loop P

2B FB 60 01 E8 03 00 00(P:1000)

Write position ring D

2B FB 60 03 E8 03 00 00(D:1000)

## PDO mapping instance:

Configure a TP DO and an RPDO to achieve PDO control motor movement

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|      | Target                     | Explain               | Index             | Subindex        |
|------|----------------------------|-----------------------|-------------------|-----------------|
| RPDO | Controlword                | control word          | 6040 <sub>h</sub> | 00 <sub>h</sub> |
|      | Modes_of_operation         | operator schema       | 6060 <sub>h</sub> | 00 <sub>h</sub> |
|      | Target_position            | target location       | 607A <sub>h</sub> | 00 <sub>h</sub> |
|      |                            |                       |                   |                 |
| TPDO | Statusword                 | status word           | 6041 <sub>h</sub> | 00 <sub>h</sub> |
|      | Modes_of_operation_display | Actual operation mode | 6061 <sub>h</sub> | 00 <sub>h</sub> |
|      | Target_position            | physical location     | 6064 <sub>h</sub> | 00 <sub>h</sub> |

Step 1: Set the mapping parameters for the RPDO and TPDO

Write the configuration and mapping parameters for RPDO / TPDO via SDO-, CS command

character: write 4 bytes

Index: PDO mapping index (where T PDO is 1A00<sub>h</sub>, An RPDO of 1600<sub>h</sub> )

Subindex: PDO parameter order, the subindex is the parameter order, the value

Value: Target index + subindex + data type (like 10004060<sub>h</sub> Represents as: index 6040<sub>h</sub>,

Subindex 00<sub>h</sub>, Length 10<sub>h</sub> )

| SDO<br>instruct | Byte0 | Byte 1 | Byte 2 | Byte 3   | Byte 4 | Byte 5 | Byte 6 | Byte 7 |
|-----------------|-------|--------|--------|----------|--------|--------|--------|--------|
|                 | CS    | Index  |        | subindex | Value  |        |        |        |
| RPDO            | 23    | 00     | 16     | 01       | 10     | 00     | 40     | 60     |
| RPDO            | 23    | 00     | 16     | 02       | 08     | 00     | 60     | 60     |
| RPDO            | 23    | 00     | 16     | 03       | 20     | 00     | 7A     | 60     |
| TPDO            | 23    | 00     | 1A     | 01       | 10     | 00     | 41     | 60     |
| TPDO            | 23    | 00     | 1A     | 02       | 08     | 00     | 61     | 60     |
| TPDO            | 23    | 00     | 1A     | 03       | 20     | 00     | 64     | 60     |

Complete the mapping

Step 2: Set TPDO and communication parameters

TPDO1 (1800<sub>h</sub>) Return a PDO after receiving two synchronization frames

Set to (send) to an asynchronous periodic type?transmission\_type (1800 h: 02 h )= FFh

Prohibited time of 2ms (20 100us)?inhibit\_time (1800h: 03 h ) = 14 h cycle time 10ms (10 1ms)

event\_time (1800 h:05 h )= 0A h

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| SDO      | Byte0 | Byte 1 | Byte 2 | Byte 3   | Byte 4 | Byte 5 | Byte 6 | Byte 7 |
|----------|-------|--------|--------|----------|--------|--------|--------|--------|
| instruct | CS    | Index  |        | Subindex | Value  |        |        |        |
| TPDO     | 23    | 00     | 18     | 02       | 02     | 00     | 00     | 00     |
|          |       |        |        |          |        |        |        |        |
|          |       |        |        |          |        |        |        |        |

The PDO mapping is complete.

### Version correction

Version and new: BUG repair description, optimization function description.

Version BUG: found version BUG, with red awake target out in the bug version source, why

#### 20230518

I. Add version record index 20FF (record value is year month date, if read 20230518 is May 18,2023)

II. Optimize save instructions, send immediately into effect, no restart

BUG: Current setting does not work (20230519 repair)

Reason: Improper adjustment of the save function caused by the function.

#### 20230519

I. Fixed 20230518 BUG with current settings take effect

II. Optimize the current position as the target position of the enabling motor to solve the problem of reenabling rotation due to the target position

III. Optimize the speed of the BUG

IV. Optimize the operation mode of 4

#### 20230525

I. Optimize control logic

II. Optimize the operation mode enabling logic, state word mode display logic

III. Multiple indexes are added for subsequent program programming implementation

Found BUG: mailbox problem PDO can send only 3 (20230627 fix)

Reason: The can mailbox is blocked and the excess data is filtered

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BUG: torque rotation direction and torque current value (20230627)

Reason: the torque rotation direction program is wrong, and the torque current value problem is the coefficient problem

## 20230627

I. Optimize the mailbox BUG

II. Optimize the rotation direction of the torque mode BUG, optimize the torque mode current value BUG,

III. The optimized current maximum current setting takes effect immediately

## 20230817 (no program modification, this is a document modification)

Re-proofread the agreement document, and update the original document with incorrect representation

BUG found: POD result of mailbox problem PDO mapping, PDO failed (20231202 fix)

Reason: canopen error state (0x603f, original 16 bit) and can error state (0x0A, 32 bit) match, general index 0x603F mapping length uses 16 bits, causing PDO configuration problem, without 603F!

## 20240122

Optimize BUG: restore 603F original index length of 16 bits and match the general error report

Optimization: Support NMT management, heartbeat message support,

Optimization: Heartbeat message missing error,

Optimization: The heartbeat time is longer than the timer time, causing the timer to overflow and stop sending

Optimization: Torque mode - Using percentage mode:

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NMT

## 1. Heartbeat message

The slave station periodically sends packets to the master station during the "monitoring time". If the master station does not receive the next heartbeat packet from the slave station after the "Heartbeat Consumer time" expires, the master station determines that the slave station is faulty.

Table 10-15 Formats of heartbeat packets

| COB-ID        | Byte 0 |
|---------------|--------|
| 0x700+Node_ID | Status |

Table 10-16 Status values

| Status value | Meaning                        |
|--------------|--------------------------------|
| 0x00         | Start(bo-tp)                   |
| 0x04         | Stop(Stopped)                  |
| 0x05         | Running(Operathional)          |
| 0x7f         | Pre-operation(Pre-operational) |

## 2. Node protection

The master station periodically sends the packet to the slave station in the "monitoring time", and the slave station responds immediately after receiving the packet. If the master station does not receive the packet from the slave station after the "Monitoring time \* Life factor" time, the master station judges that the slave station is wrong!

Master request message format -- (0x700+ node number) (the message has no data)

Slave response message format -- (0x700+ node number) + status

Table 10-17 Reply packets from the slave station

| COB-ID        | Byte 0                                     |
|---------------|--------------------------------------------|
| 0x700+Node_ID | Bit7:Trigger Position<br>Bi16-Bt10: status |

Table 10-18 Description of the slave station reply packet status values

| Status value | Meaning                  |
|--------------|--------------------------|
| 0            | Initialize(Initializing) |

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|     |                                 |
|-----|---------------------------------|
| 1   | Dis-connct (Disconnected)       |
| 2   | Connect (Counogting)            |
| 3   | Prepare (Prepare)               |
| 4   | Stop(Stopped)                   |
| 5   | Running(Operational)            |
| 127 | Pre-operation (Pre-operational) |

Status - The data section contains a trigger bit (bit7). The trigger bit must be set to 0 or 1 in each node protection response, and the trigger bit must be set to 0 in the first node protection request. 6(bit0-6) indicates the node status.

The standard CAN slave station generally supports only one type of node protection, and the FD3 series servo drive supports both types of protection. However, a node cannot support both node protection and heartbeat packets at the same time. Only one of them can be used for protection.

### Startup procedure description

CANopen supports both extended boot-up and minimal boot-up procedures during network initialization. This initialization process can be represented by a nodal state transition diagram, as shown in Figure 10-7.

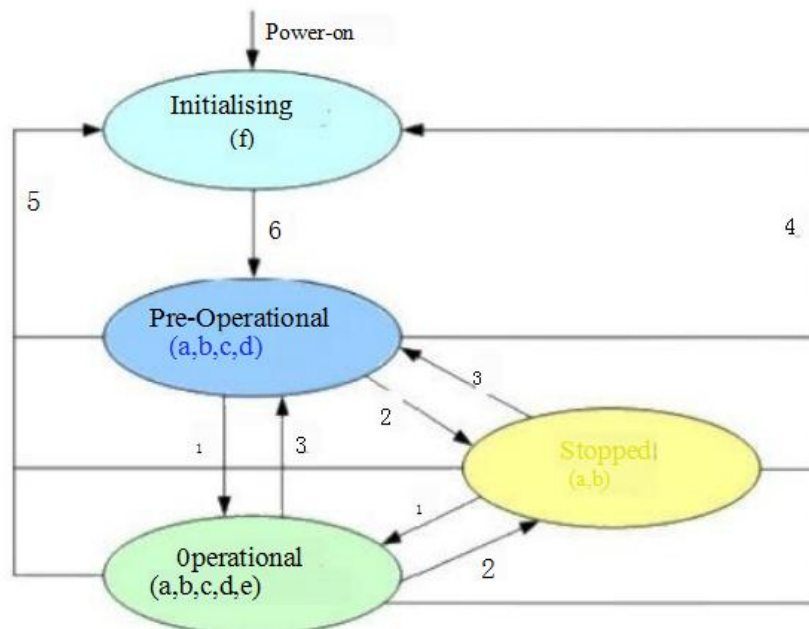


Figure 10-7 Node status change

Note: The letters in parentheses in the figure indicate communication objects that can be used in



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different states.

therein : a—NMT                      d—Emergency                      b—Node Guard  
e—PDO                      c—SDO                      f—Boot-up

NMT management messages can be used to switch between modes. Only the NMT-Master node can send NHT Module Control messages. All slave devices must support the NMT Module Control service, and NMT Module control messages do not need to be answered.

After the initialization, the device automatically enters the Pre\_Operational state and sends Boot-up messages. The NMT message format is as follows :

NMT Master—NMT Slave(s)

Table 10-19 Formats of NMT management packets

| COB-ID | Byte | Byte1   |
|--------|------|---------|
| 0x000  | CS   | Node-ID |

When Node-ID=0, it indicates that all NMT slave devices are addressed. CS is the command word, whose values are shown in Table 10-15

Table 10-15 Values of CS

|      |                                 |
|------|---------------------------------|
| 0x01 | Enable remote node              |
| 0x02 | Close remote node               |
| 0x80 | Enter the pre-buttoned register |
| 0x81 | Reset node                      |
| 0x82 | Reset communication             |